



Cloud Computing Fundamentals

Agenda

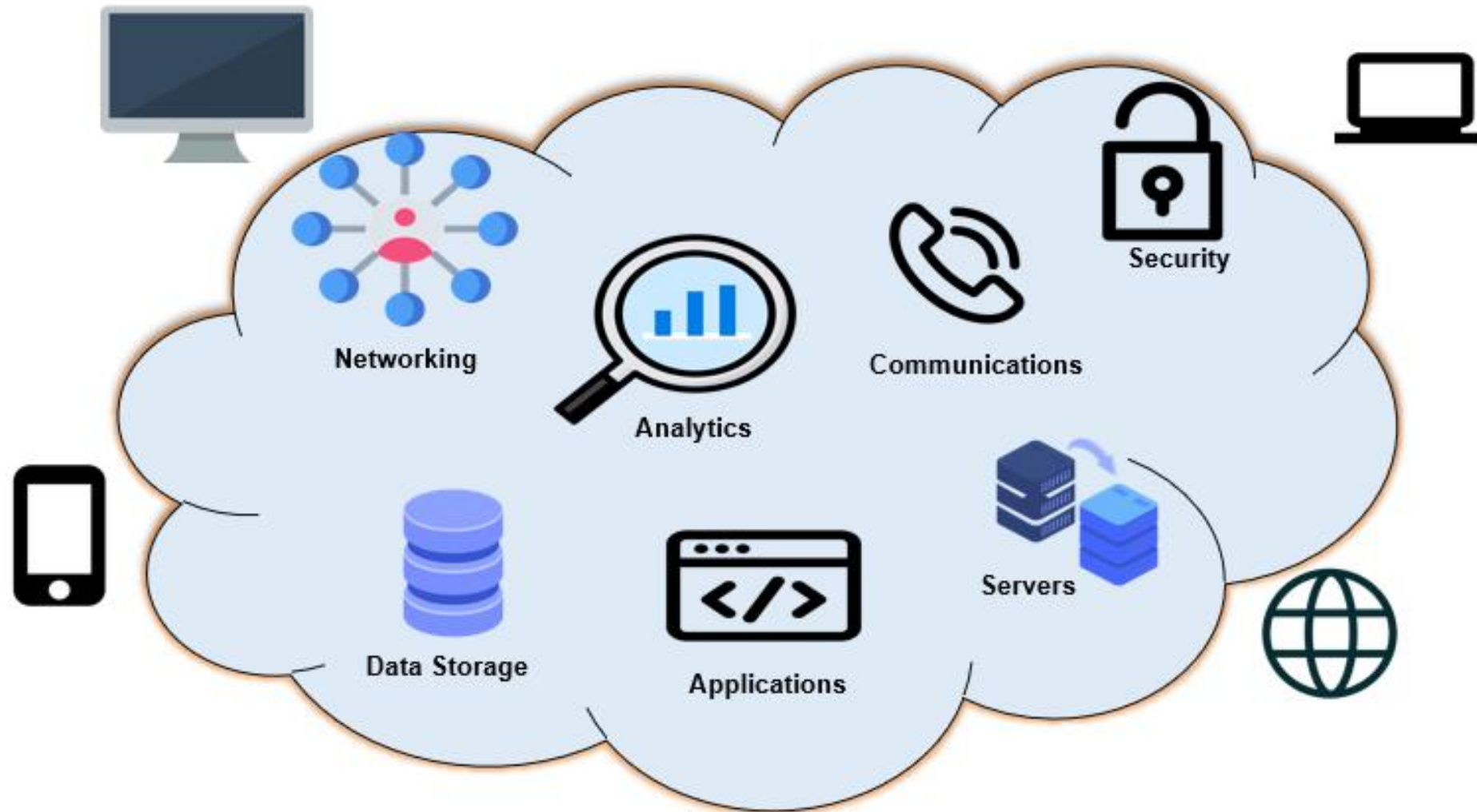
Overview

Benefits of Cloud Computing

Types of Cloud Services

Shared Responsibility

Introduction of Cloud Computing



Overview

- Microsoft cloud offerings enable organizations to eliminate computing infrastructure that they might normally host themselves.
- Larger organizations typically host their own servers, networking equipment, and other IT resources in a *data center*, which is a facility specifically designed and constructed to house servers and other IT hardware and related infrastructure.



Overview

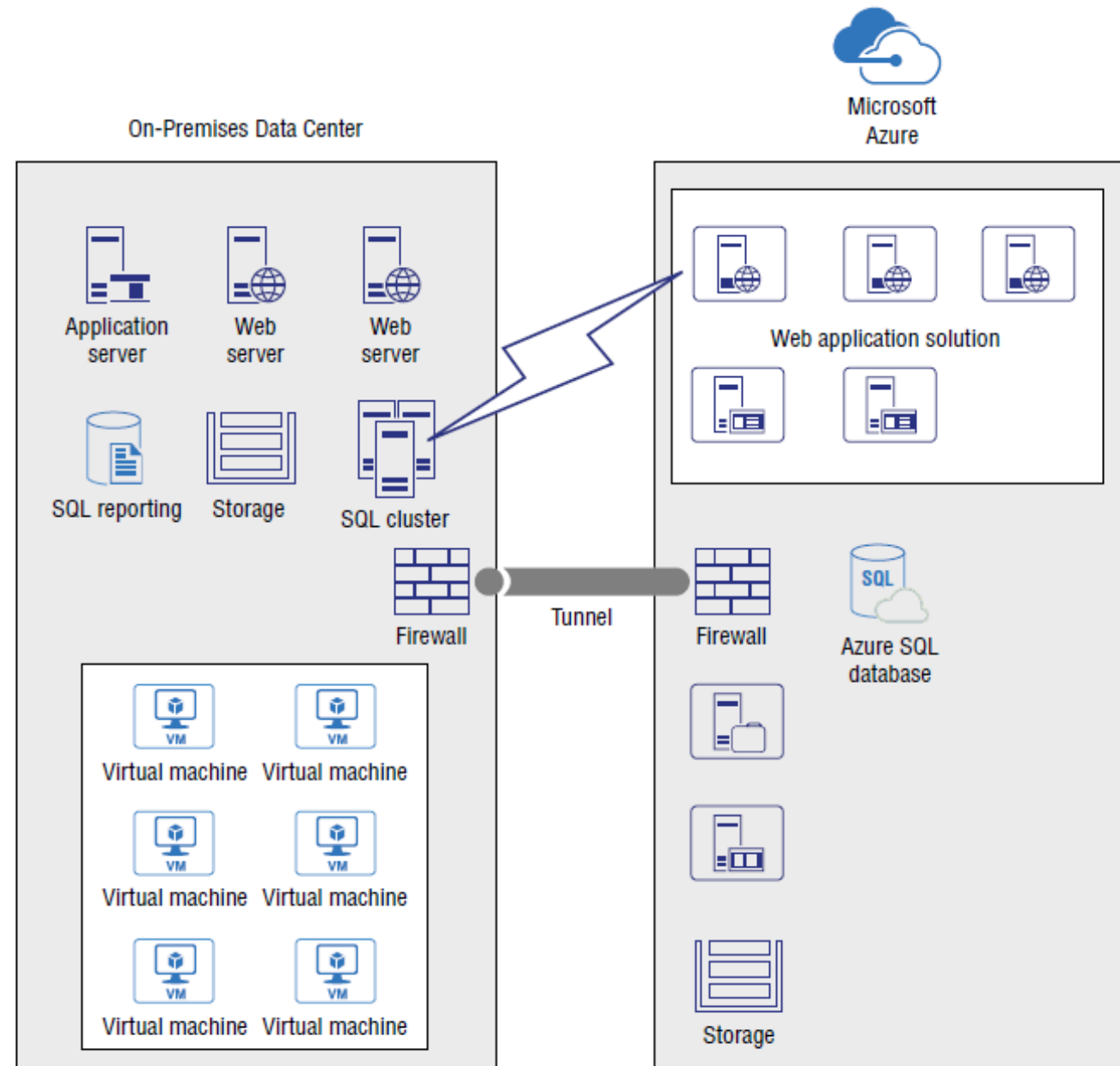
- Smaller organizations generally either use a third-party data center or place their servers and other IT infrastructure in one or more *server rooms*, which are essentially very small data centers housed inside the company's facility.



Overview

- A cloud offering such as Azure enables organizations to move some—if not all—of their servers, networking equipment, and other IT resources into a data center managed by another company.
- In the case of Azure, Microsoft owns and maintains numerous data centers around the world to host these resources for all sizes of organizations.
- Managing these resources then becomes a shared responsibility between the organization and Microsoft.

Overview



Benefits of Cloud Computing

- Economic Benefits
- Scalability and Elasticity
- High Availability
- Fault Tolerance
- Disaster Recovery
- Human Resources

Economic Benefits

- With Azure, Microsoft handles the capital expenditures necessary to maintain and grow the service.
- An organization using Azure services therefore eliminates those capital expenditures and replaces them with *operational expenditures (OpEx)*, which are monthly expenditures that the organization uses to run its operation.
- Another economic benefit to cloud computing is *economy of scale*, in which a cloud provider can purchase large amounts of hardware at a discount and pass that discount along to its customers.

Consumption-Based model

- Cloud computing falls under OpEx because cloud computing operates on a consumption-based model.
- With cloud computing, you don't pay for the physical infrastructure, the electricity, the security, or anything else associated with maintaining a datacenter.
- Instead, you pay for the IT resources you use. If you don't use any IT resources this month, you don't pay for any IT resources.

Consumption-Based model

- No upfront costs.
- No need to purchase and manage costly infrastructure that users might not use to its fullest potential.
- The ability to pay for more resources when they're needed.
- The ability to stop paying for resources that are no longer needed.

Scalability and Elasticity

- Scalability is the ability to add computing resources to adjust to increased demand.
- Scaling generally comes in two varieties: vertical and horizontal.
- Vertical scaling is focused on increasing or decreasing the capabilities of resources - you could vertically scale up to add more CPUs or RAM to the virtual machine.
- Horizontal scaling is adding or subtracting the number of resources - you could add additional virtual machines .

Scalability and Elasticity

- You can *scale out* your servers, adding additional ones to meet the increased demand, then *scale in* when the peak season is over - horizontal scaling.
- *Scaling up*, also called *vertical scaling*, refers to adjusting capacity like memory in existing resources to accommodate demand changes.
- When you remove the extra memory when you no longer need it, you're *scaling down*. - vertical scaling
- Automatic scaling, called *elasticity*, enables Azure to scale resources for you without interaction.

High Availability

- High availability (HA) describes a system that is available for use without significant outages and that is generally backed by a *service level agreement (SLA)*.
- For example, if a service has an SLA of 99.9 percent, the service is guaranteed to be available 99.9 percent of the time.
- Translated to the real world, that means the service can be unavailable no more than 43.2 minutes in a 30-day period to meet the 99.9 percent SLA for that month.
- <https://uptime.is/>

Fault Tolerance

- The term *fault tolerance* describes a characteristic of a system that enables it to continue functioning when one or more components of the system fails.
- If the web server goes down, the service becomes unavailable. To make it fault tolerant, you can add a second web server and balance traffic between them.

Disaster Recovery

- *Disaster recovery* refers to the process of recovering from a situation where multiple systems or services fail.
- For example, assume that your company's primary data center is hit by a tornado, destroying all the IT infrastructure and services hosted at that location.
- Azure has complex disaster recovery plan that can include multiple data centers, active mirroring of services between data centers.

Human Resources

- The indirect consequence of outsourcing IT infrastructure and services to a cloud provider like Microsoft always has some effect on the IT staff who are managing the infrastructure and services that move to the cloud.

Cloud Computing Models – IaaS, PaaS, SaaS

Round 1: SaaS (You just eat pizza)

Scenario:

You order from Zomato

What you do:

- Choose pizza
- Pay
- Eat

What you DON'T care about:

- Kitchen setup
- Chef
- Ingredients
- Oven

Mapping:

SaaS = Ready-to-use software

Azure Example:

- Microsoft 365
- Gmail

Cloud Computing Models – IaaS, PaaS, SaaS

🍕 Round 2: PaaS (You cook, platform is ready)

Scenario:

You get a **fully equipped kitchen**

- Oven ready
- Ingredients available
- Gas, electricity handled

👉 What you do:

- Cook pizza
- Decide recipe

👉 What you DON'T manage:

- Infrastructure
- Hardware
- OS updates

💡 Mapping:

PaaS = You focus on code, Azure handles platform

💬 Azure Example:

- Azure App Service
- Azure SQL Database

Cloud Computing Models – IaaS, PaaS, SaaS

Round 3: IaaS (You build everything)

Scenario:

You get:

- Empty land
- Raw materials

What you do:

- Build kitchen
- Install oven
- Hire staff
- Cook pizza

You control EVERYTHING:

- Machines
- Storage
- Network
- OS

Mapping:

IaaS = Full control, more responsibility

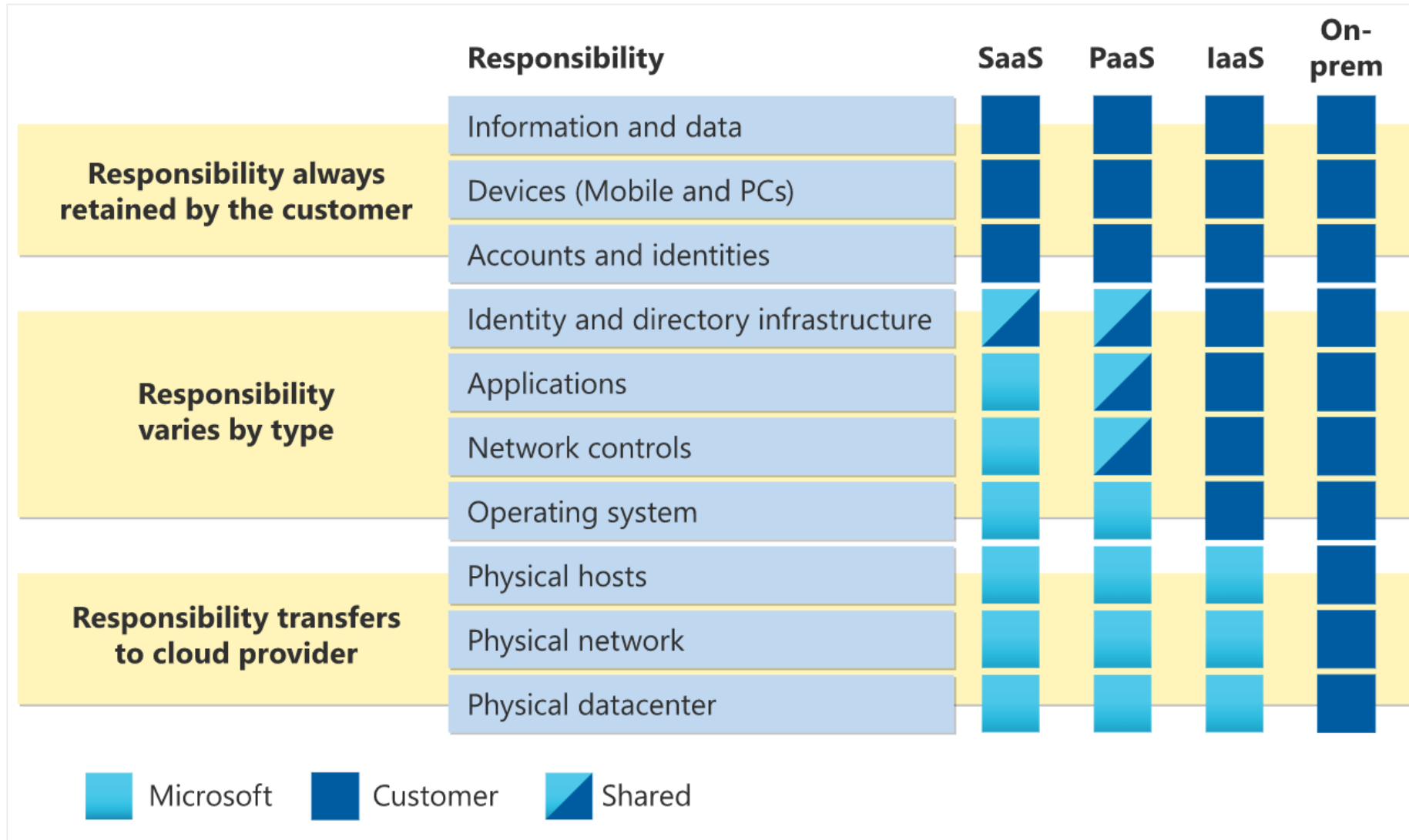
Azure Example:

- Azure Virtual Machines

Cloud Computing Models - IaaS

- Infrastructure as a service (IaaS) is the most flexible category of cloud services, as it provides you the maximum amount of control for your cloud resources. In an IaaS model, the cloud provider is responsible for maintaining the hardware, network connectivity (to the internet), and physical security. You're responsible for everything else: operating system installation, configuration, and maintenance; network configuration; database and storage configuration; and so on.
- IaaS enables you to deploy and use infrastructure components in Azure. A primary example of IaaS is *virtualization*, in which you deploy virtual servers in Azure rather than physical servers in your own data center

Cloud Computing Models



Cloud Computing Models - IaaS

Some common scenarios where IaaS might make sense include:

- Lift-and-shift migration: You're standing up cloud resources similar to your on-prem datacenter, and then simply moving the things running on-prem to running on the IaaS infrastructure.
- Testing and development: You have established configurations for development and test environments that you need to rapidly replicate. You can stand up or shut down the different environments rapidly with an IaaS structure, while maintaining complete control.

Cloud Computing Models - PaaS

- PaaS - think of PaaS as providing the capability to quickly develop and deploy web-based applications without the need to manage the underlying servers, operating systems, and other resources that are part of the developed application.

Cloud Computing Models

Some common scenarios where PaaS might make sense include:

- Development framework: PaaS provides a framework that developers can build upon to develop or customize cloud-based applications.
- Like - database management systems (Azure SQL Database), App Service
- Analytics or business intelligence: Tools provided as a service with PaaS allow organizations to analyze and mine their data, finding insights and patterns and predicting outcomes to improve forecasting, product design decisions, investment returns, and other business decisions.

Cloud Computing Models

- SAAS - SaaS is a licensing model in which an organization licenses software on a subscription basis from a provider that centrally manages and distributes the software. A perfect example of SaaS is Microsoft 365, where Microsoft provides Office applications (Word, Excel, and others) on a subscription basis.
- With SaaS, you're essentially renting or using a fully developed application. Email, financial software, messaging applications, and connectivity software are all common examples of a SaaS implementation.

Check your knowledge

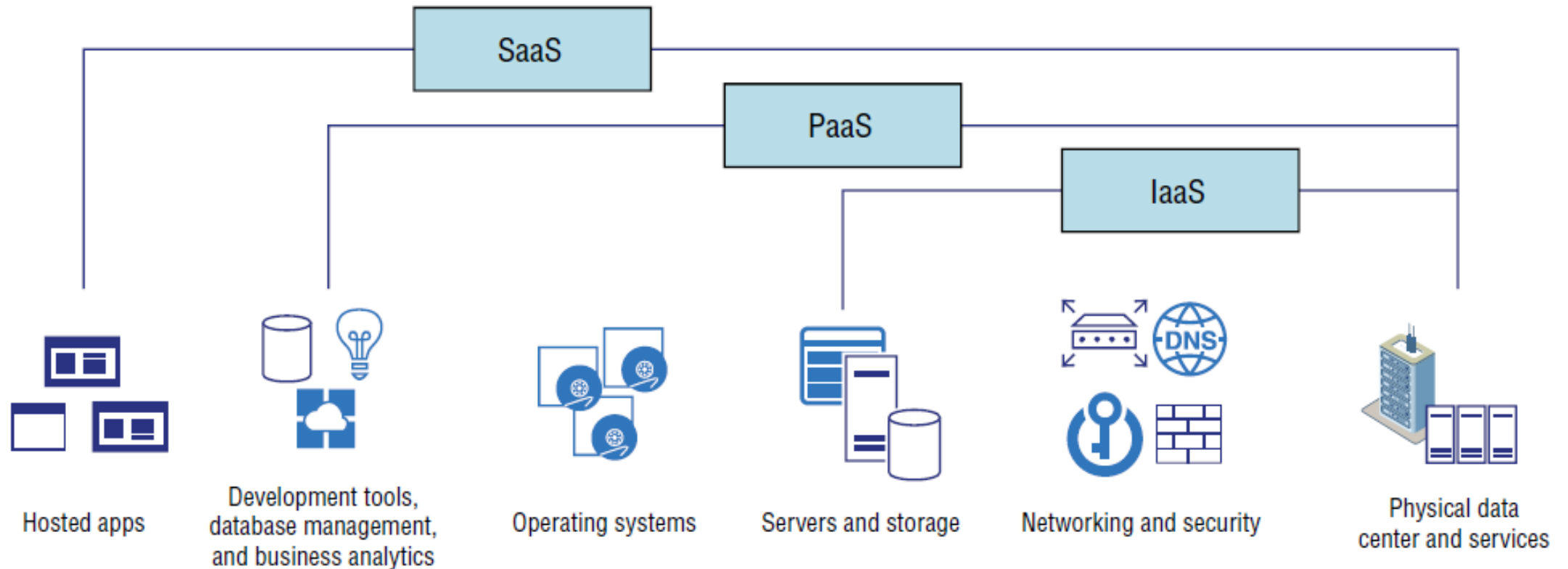
1. Which cloud service type is most suited to a lift and shift migration from an on-premises datacenter to a cloud deployment?

- ☐ Infrastructure as a Service (IaaS)
- ☐ Platform as a Service (PaaS)
- ☐ Software as a Service (SaaS)

2. What type of cloud service type would a Finance and Expense tracking solution typically be in?

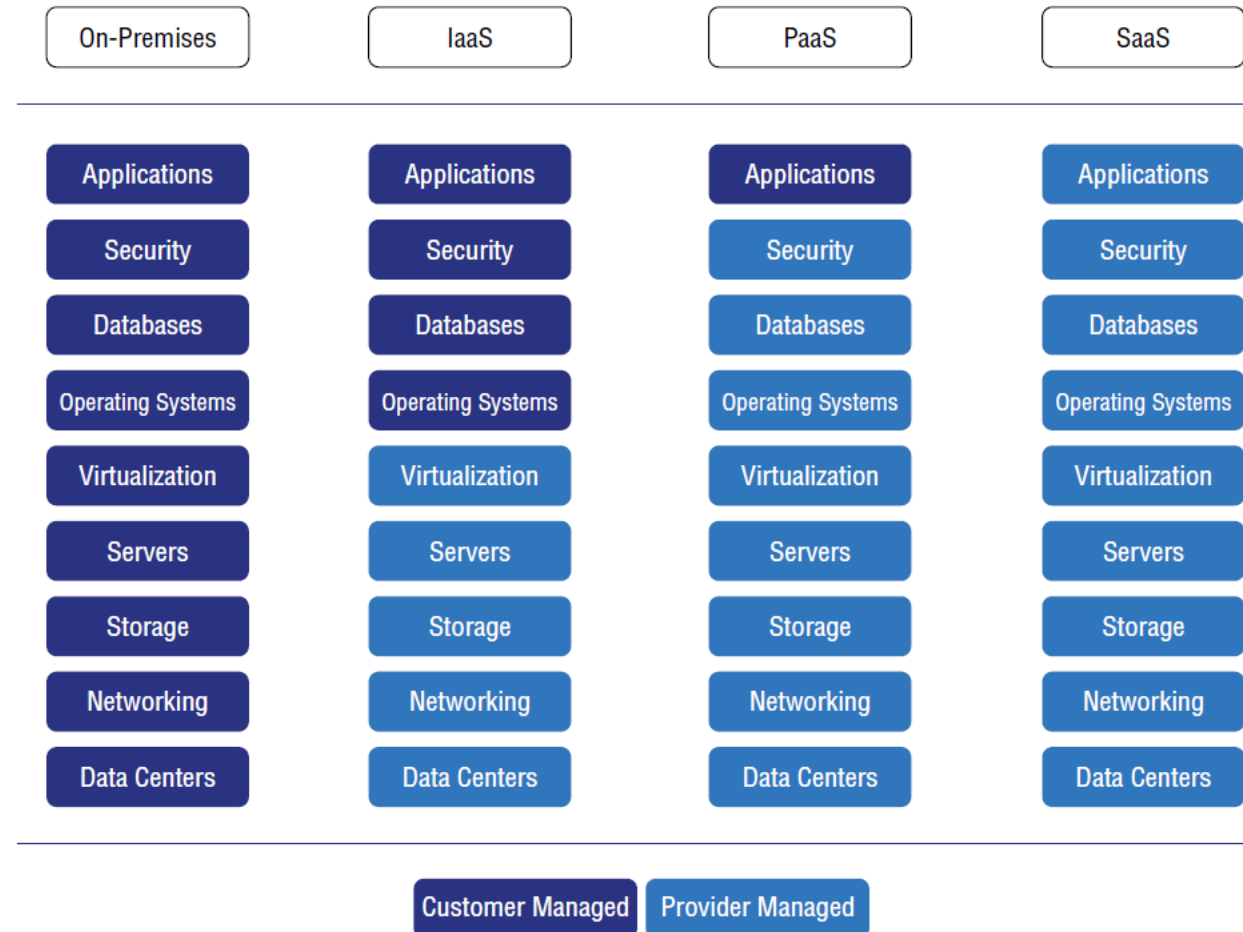
- ☐ Infrastructure as a Service (IaaS)
- ☐ Platform as a Service (PaaS)
- ☐ Software as a Service (SaaS)

Cloud Computing Models



Shared Responsibility Model

- With cloud computing, you have a shared responsibility with the cloud provider.



Shared Responsibility Model

- With the shared responsibility model, these responsibilities get shared between the cloud provider and the consumer. Physical security, power, cooling, and network connectivity are the responsibility of the cloud provider. The consumer isn't collocated with the datacenter, so it wouldn't make sense for the consumer to have any of those responsibilities.
- At the same time, the consumer is responsible for the data and information stored in the cloud. (You wouldn't want the cloud provider to be able to read your information.) The consumer is also responsible for access security, meaning you only give access to those who need it.

Shared Responsibility Model

- You'll always be responsible for:
 - The information and data stored in the cloud
 - Devices that are allowed to connect to your cloud (cell phones, computers, and so on)
 - The accounts and identities of the people, services, and devices within your organization
- The cloud provider is always responsible for:
 - The physical datacenter
 - The physical network
 - The physical hosts
- Your service model will determine responsibility for things like:
 - Operating systems
 - Network controls
 - Applications
 - Identity and infrastructure

Check your knowledge

1. Which type of scaling involves adding or removing resources (such as virtual machines or containers) to meet demand?

- ☐ Vertical scaling
- ☐ Horizontal scaling
- ☐ Direct scaling

2. What is characterized as the ability of a system to recover from failures and continue to function?

- ☐ Reliability
- ☐ Predictability
- ☐ Scalability

Cloud Models

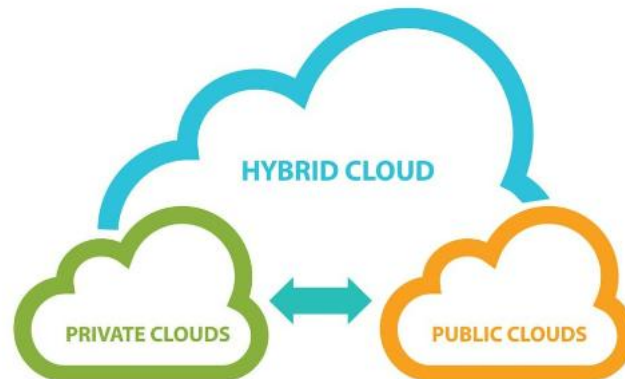
- The three main cloud models are: private, public, and hybrid.
- A private cloud is, in some ways, the natural evolution from a corporate datacenter. It's a cloud (delivering IT services over the internet) that's used by a single entity. Private cloud provides much greater control for the company and its IT department. However, it also comes with greater cost and fewer of the benefits of a public cloud deployment.

Cloud Models

- A public cloud is built, controlled, and maintained by a third-party cloud provider. With a public cloud, anyone that wants to purchase cloud services can access and use resources.

Cloud Models

- A public cloud is one in which the services offered in the cloud are available for public use over the Internet.
- A private cloud is one in which the cloud serves a single organization, whether hosted in your own data center or by someone else.
- A hybrid cloud is one in which an organization integrates on-premises services with cloud-hosted services, whether in a public or a private cloud.



Cloud Models

Public cloud	Private cloud	Hybrid cloud
No capital expenditures to scale up	Organizations have complete control over resources and security	Provides the most flexibility
Applications can be quickly provisioned and deprovisioned	Data is not collocated with other organizations' data	Organizations determine where to run their applications
Organizations pay only for what they use	Hardware must be purchased for startup and maintenance	Organizations control security, compliance, or legal requirements
Organizations don't have complete control over resources and security	Organizations are responsible for hardware maintenance and updates	

Check Your Knowledge

1. What is cloud computing?

- ☐ Delivery of computing services over the internet.
- ☐ Delivery of storage services over the internet.
- ☐ Delivery of websites accessible via the internet.

2. Which cloud model uses some datacenters focused on providing cloud services to anyone that wants them, and some data centers that are focused on a single customer?

- ☐ Public cloud
- ☐ Hybrid cloud
- ☐ Multi-cloud

3. According to the shared responsibility model, which cloud service type places the most responsibility on the customer?

- ☐ Infrastructure as a Service (IaaS)
- ☐ Software as a Service (SaaS)
- ☐ Platform as a Service (PaaS)

Azure Physical Infrastructure

Agenda

Describe the benefits and usage of Regions and Region Pairs

Describe the benefits and usage of Availability Zones

Describe the benefits and usage of Resource Groups

Describe the benefits and usage of Subscriptions

Why Azure?

- While there are many cloud computing platforms available, two platforms dominate the cloud computing industry.
- Amazon Web Services (AWS) and Microsoft Azure are the two giants in the world of cloud computing.
- While AWS is the largest cloud computing platform, Microsoft Azure is the fastest-growing and second-largest.
- AWS is up to 5 times more expensive than Azure for Windows Server and SQL Server.

Azure Data Center

- Physical facility – 200+
- Hosting for groups of networked servers
- Own Power, Cooling and Network infrastructure

Azure Data Center Region	Location
Central India	Pune
South India	Chennai
West India	Mumbai

Data Centers

Number of data centres across the world



Azure Region

- Group of Data Centers
- Different regions of many sizes around the globe
- They can contain one or more data centers
- Connected with low latency network
- Location for your service
- Services may not be available in all regions
- Azure has 60+ regions

Azure Sovereign Regions

- Sovereign regions are instances for compliance or legal purposes.
- US DoD Central, US Gov Virginia, US Gov Iowa and more - These regions are physical and logical network-isolated instances of Azure for U.S. government agencies and partners.

Azure Region Pair

- A region pair consists of two regions within the same geography. Updates are rolled out to regions serially, meaning only one region is updated at a time.
- An issue with an update or an outage would then affect only one region in the pair.
- **Resiliency** is a system's ability to recover from failures and continue to function and minimizes downtime or data loss.

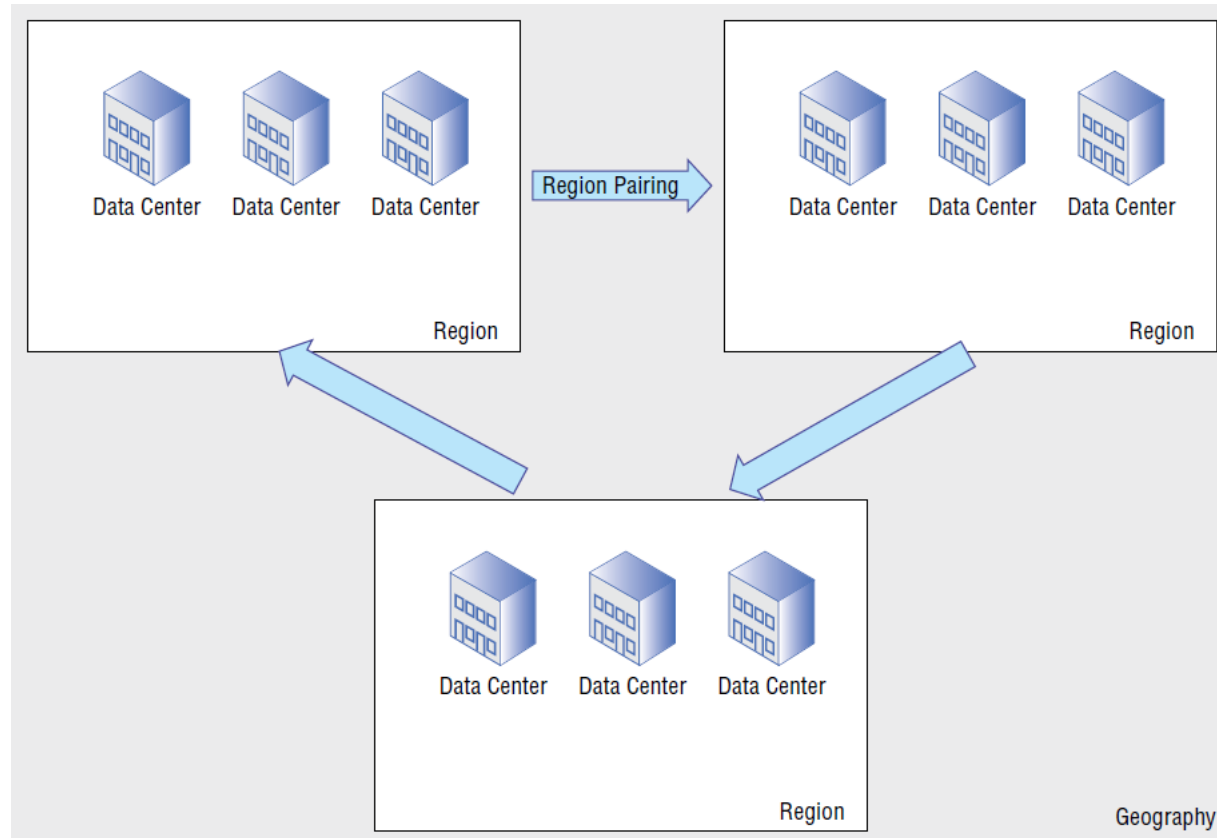
Core Azure Architectural Components

- Azure is a distributed cloud offering with data centers located in many geographical regions across the world.
- The Azure geographies generally align to specific countries such as the United States, Canada, Australia, and so on.
- However, Azure geographies can also be aligned to specific markets, such as Europe and Asia.
- Within each geography are *Azure regions*.
- A region is a grouping of data centers that interact to provide redundancy and availability for the services hosted within that region.

Core Azure Architectural Components

- For example, West US, Central US, and North Central US are three of many regions in the United States.
- Each region is paired with another in the same geography to allow for replication of resources across multiple data centers to reduce the effects of natural disasters, outages, or other potential events that would affect a given data center's ability to serve up the services hosted in that data center.
- For example, West US and East US are paired regions, North Europe and West Europe are paired, and UK West and UK South are paired.

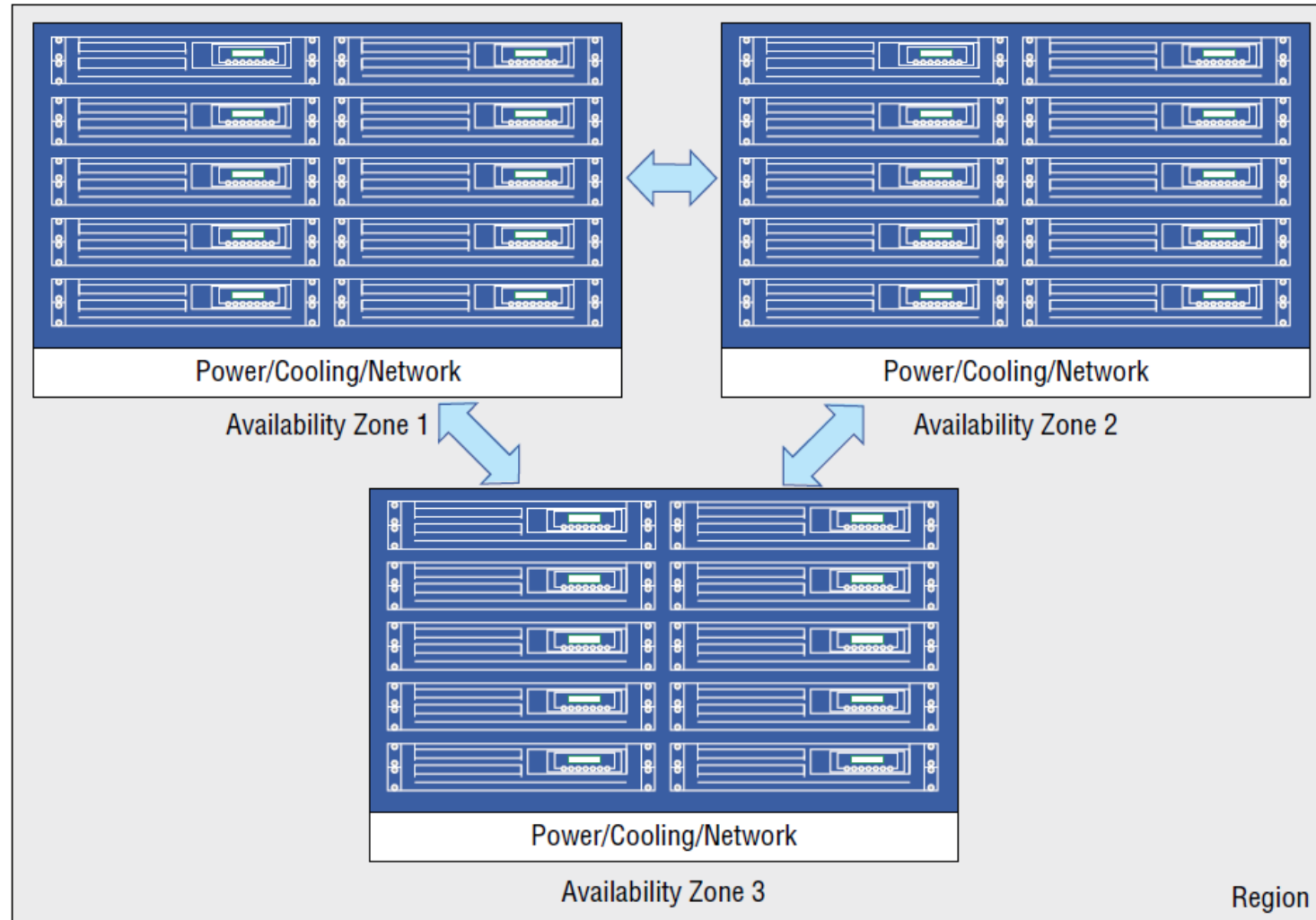
Geographies and Regions



Availability Zones

- Azure offers another level of availability protection through *availability zones*. An availability zone is a physically separate zone within a region, each with its own power, network, and cooling.
- An availability zone might encompass more than one data center. There are a minimum of three availability zones per region
- You deploy VMs to an additional availability zone so that if an incident does occur at one of the data centers, the VMs in the other availability zone will be unaffected.

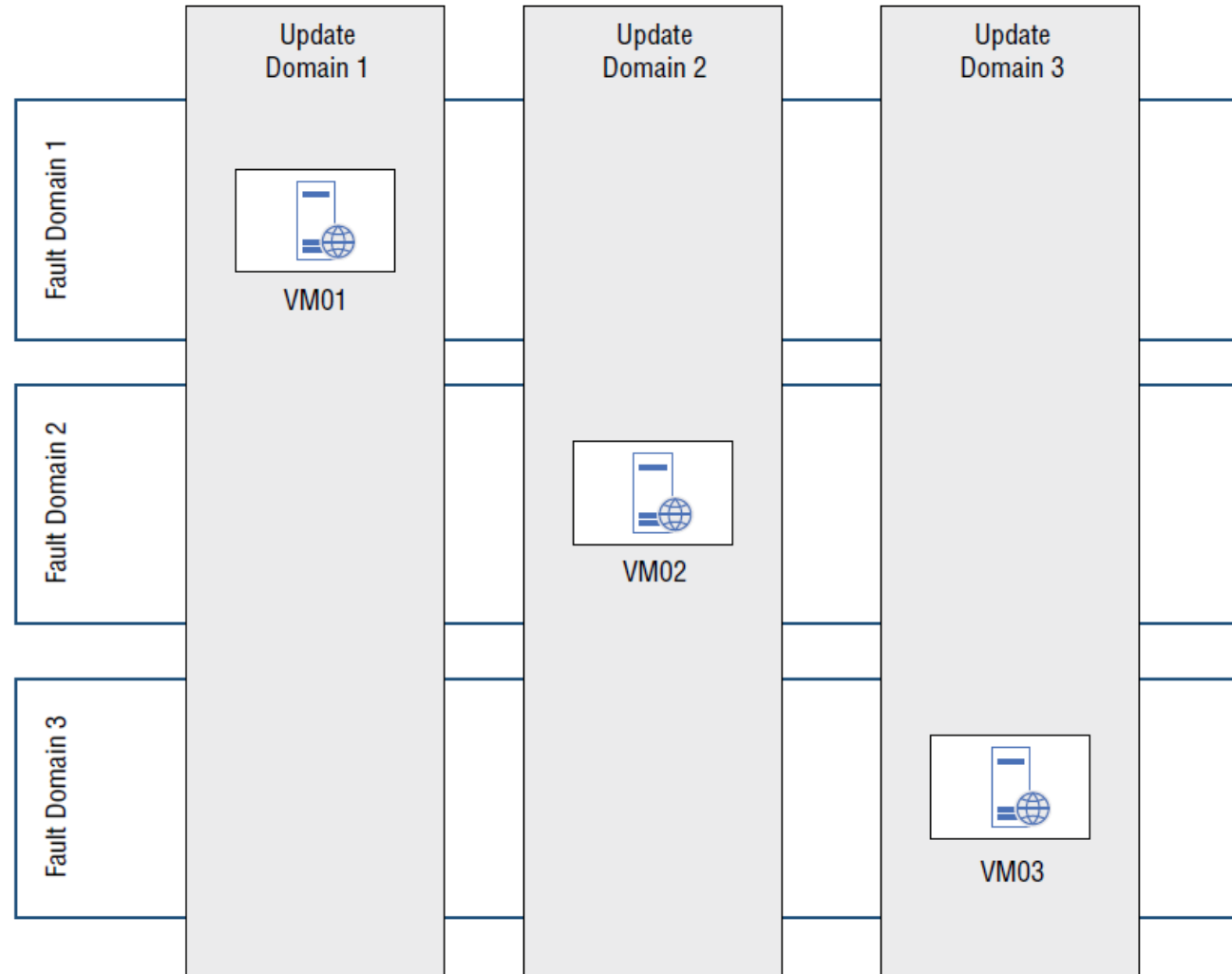
Availability Zones



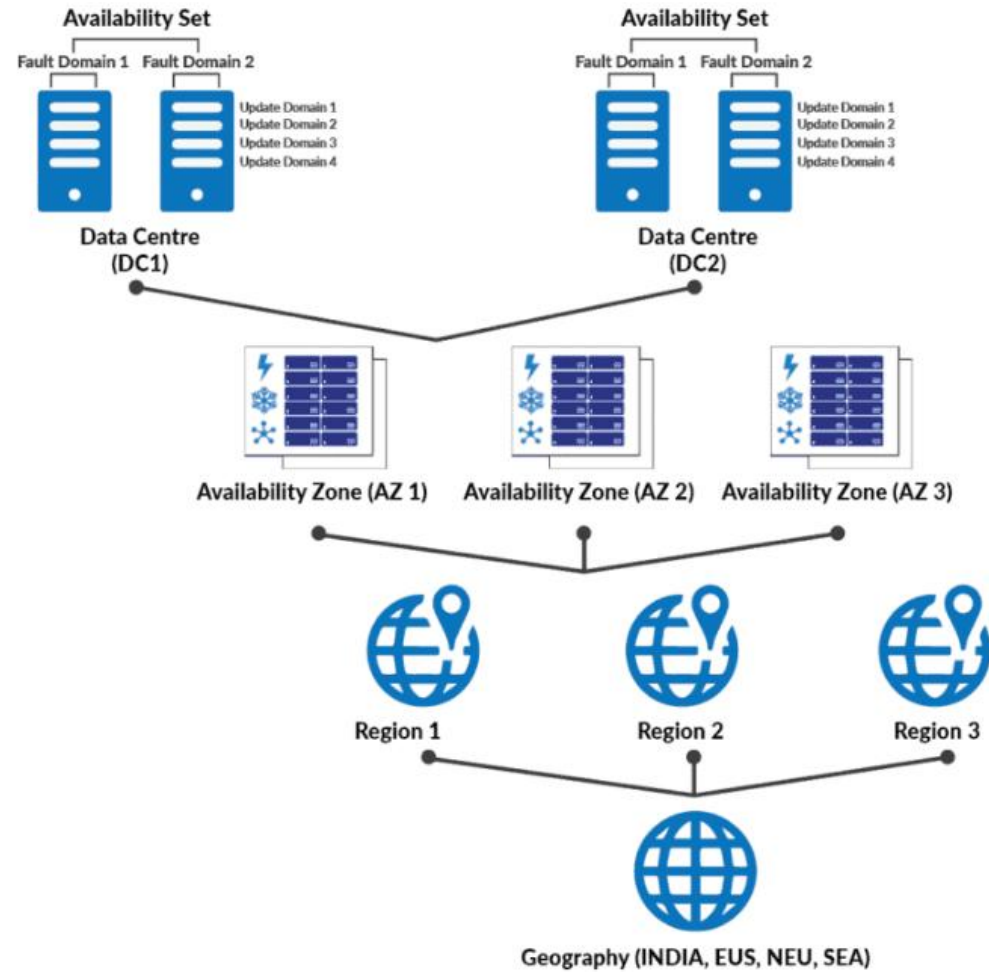
Availability Sets

- Two elements that enable availability sets are *update domains* and *fault domains*. A fault domain is a logical grouping of hardware that shares a power source and network switch, similar to a physical rack in a data center.
- An availability set distributes VMs across multiple fault domains and update domains

Availability Sets



Infra Architecture



Check your knowledge

1. How many resource groups can a resource be in at the same time?

- ☐ One
- ☐ Two
- ☐ Three

2. What happens to the resources within a resource group when an action or setting at the Resource Group level is applied?

- ☐ Current resources inherit the setting, but future resources don't.
- ☐ Future resources inherit the setting, but current ones don't.
- ☐ The setting is applied to current and future resources.

3. What Azure feature replicates resources across regions that are at least 300 miles away from each other?

- ☐ Region pairs
- ☐ Availability Zones
- ☐ Sovereign regions

Infra Architecture

- <https://azure.microsoft.com/en-us/explore/global-infrastructure/>
- <https://infrastructuremap.microsoft.com/>

Azure Resources and Resource Groups

Agenda

Resources

Resource Groups

Subscription

Billing

Core Azure Services

Resources

- Azure *resources* are manageable items in Azure such as virtual machines, databases, virtual networks, and storage accounts.
- If you deploy only a few resources to Azure, managing them will likely not be a significant challenge. However, your Azure solution may grow to encompass a very large number of resources of different types, all potentially scattered across multiple regions.
- As the number of resources grows, your ability to manage them individually quickly becomes difficult.

Resource Groups

- *Resource groups* provide a means for organizing and managing resources.
- Think of a resource group as a **logical container** for one or more resources.
- You can apply various properties to the resource group, and those properties apply to all the resources in that resource group.
- For example, assume you have a group of resources that you want to prevent from being deleted but want resource admins to be able to modify. You can create a resource group, assign the resources to that group, and apply a CanNotDelete lock on the resource group. Any resources that you later apply to the resource group will also be automatically protected from deletion.

Resource Groups

- You can use resource groups to apply policies to a group of resources, control who has administrative permissions for the resources in the group, and perform other operations on the resources as a single entity.
- For example, assume you stand up 10 VMs for a proof-of-concept project. When the project is completed and it's time to delete the VMs, you simply delete the resource group.

Resource Groups

- A resource can exist in only one group, but you can add or remove a resource to or from the group as needed. You can also move resources from one group to another.
- Resources in a group can reside in various regions
- Resources in different groups can interact with one another
- The resource group stores metadata about the resources. Therefore, when you specify a location for the resource group, you are specifying where that metadata is stored.

Azure Resource Manager (ARM)

- ARM is the service that enables you to manage resources.
- ARM is not a tool or interface.
- When you create a VM in the Azure portal, the portal sends the properties to the ARM application programming interface (API). ARM then communicates with the resource provider to create the VM.
- ARM supports the use of templates for creating, managing, and monitoring resources.
- ARM templates are JavaScript Object Notation (JSON) files that define one or more resources

Subscription

- Using Azure requires an Azure subscription. A subscription provides you with authenticated and authorized access to Azure products and services.
- An Azure subscription links to an Azure account, which is an identity in Azure Active Directory (Azure AD) or in a directory that Azure AD trusts.
- An account can have multiple subscriptions, but it's only required to have one.

Subscription

- Similar to using resource groups to separate resources by function or access, you might want to create additional subscriptions for resource or billing management purposes.
 - Environments: You can choose to create subscriptions to set up separate environments for development and testing
 - Organizational structures: You can create subscriptions to reflect different organizational structures.
 - Billing: You can create additional subscriptions for billing purposes. Because costs are first aggregated at the subscription level

Subscription

- Azure subscription serves as a logical container for your Azure resources but at a higher level.
- Think of a subscription as a big box that contains all your resource group boxes.
- Also, a resource group can only exist in one subscription.
- In a simple Azure environment, you will likely have a single Azure subscription that contains all your Azure resources. As the complexity of your environment grows, however, you might want to use multiple Azure subscriptions to simplify resource management, billing, and cost containment.

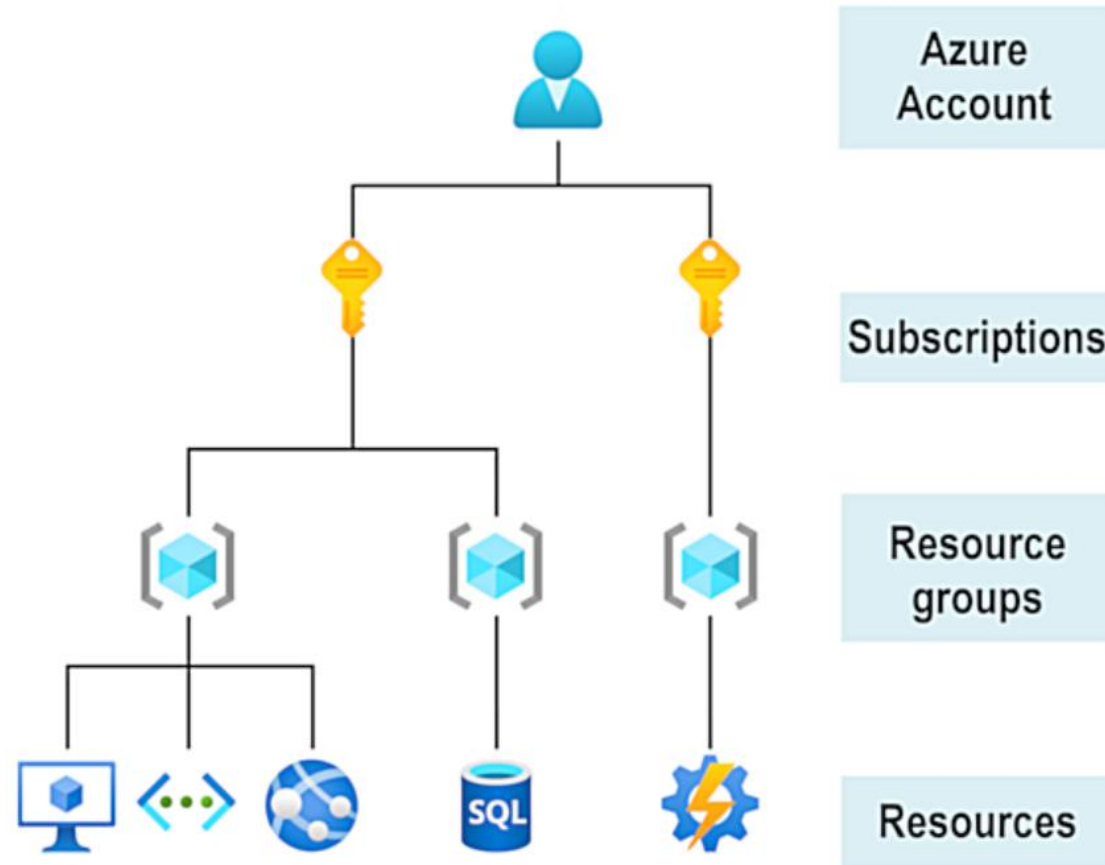
Subscription

- Azure subscriptions also represent a payment agreement.
- If you choose a pay-as-you-go offer tied to a credit card, for example, the resulting subscription includes the credit card number and related billing information required to bill your Azure consumption each month to that credit card.
- Free, pay-as-you-go, and member offers are the three primary types of subscriptions

Azure Billing Accounts

- A *billing account* is a mechanism that you use to pay for Azure services.
- Microsoft Online Services Program - This type of billing account is added when you sign up for Azure services through the Azure website, such as an Azure Free Account, an account with pay-as-you-go billing
- Enterprise agreement – your organization might purchase on-premises licenses
- Microsoft Customer Agreement

Azure Accounts

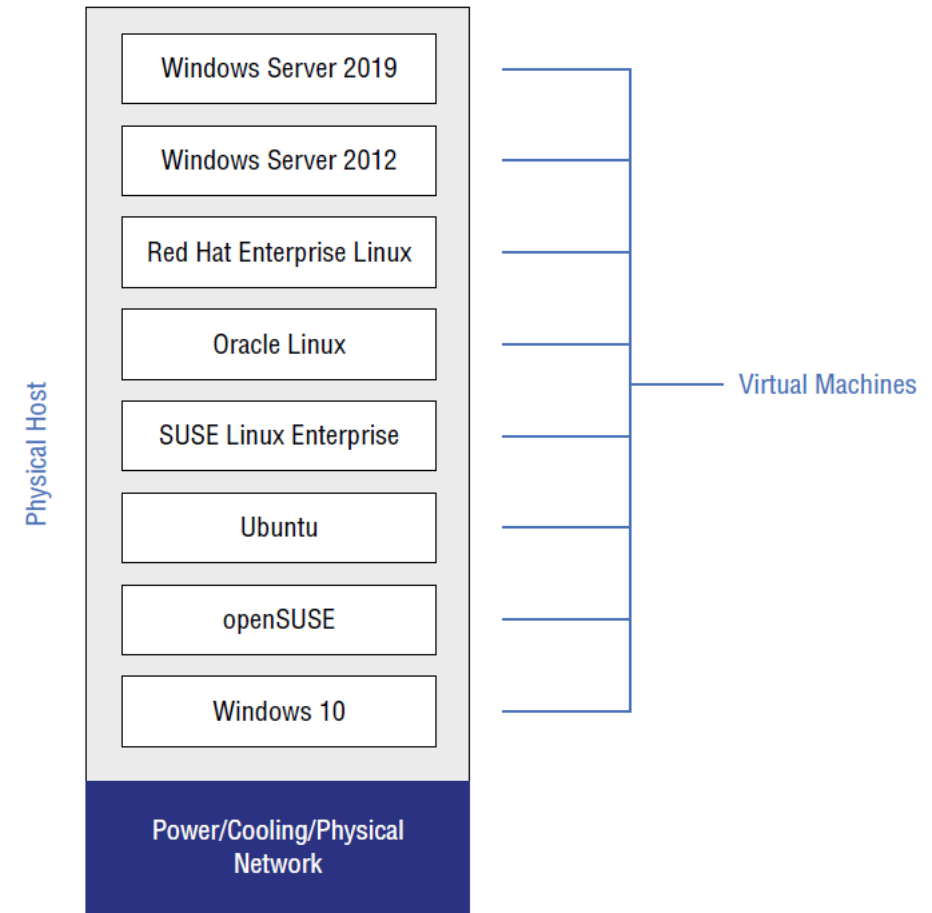


Core Azure Services

- Virtual Machines
- App Services
- Azure Container Instance
- Azure Kubernetes Service
- Azure Storage
- Azure Data Service

Virtual Machines

- Although physical computers are still a major component of many organizations' IT infrastructure, organizations are increasingly moving toward the use of *virtual machines* (VMs).
- The VM still runs on a physical device generally referred to as a *host*. The VM is a *guest* on that host.



Virtual Machines

- Virtual machines offer many advantages over physical machines. First, they reduce power and physical space requirements because multiple VMs can run on a single host.
- For example, the data center rack space needed to host 10 physical servers might host 100 or more virtual servers.
- VMs also offer considerable flexibility over physical machines. Because VMs are software based, you can create them very quickly by specifying their parameters and handing the actual creation task over to the hypervisor.

Virtual Machines

- Scalability is another advantage to using VMs. You can easily add resources like CPUs, cores, and memory to a VM by modifying its configuration settings.
- VMs are not limited to running whatever operating system is deployed on the host.
- Finally, VMs can be moved easily from one host to another because they are software based.

Virtual Machines

- Just like a physical computer in your datacenter, VMs have several elements that are needed to do their job:
 - The VM itself
 - Disks for storage
 - Virtual network
 - Network interface to communicate on the network
 - Network Security Group (NSG) to secure the network traffic
 - An IP address (public, private, or both)

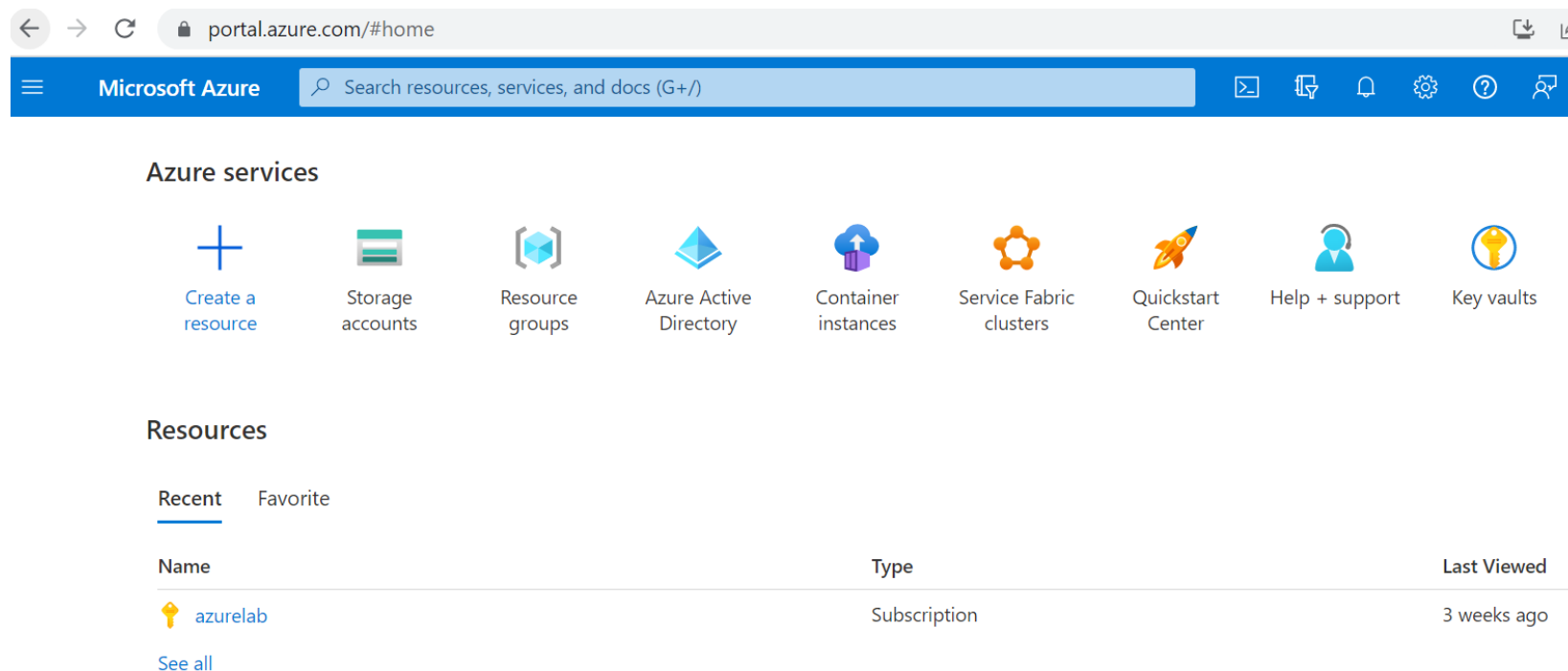
Virtual Machines

checklist of things to think about

- The network
- VM name
- Location
- The VM size
- Disks
- Operating system

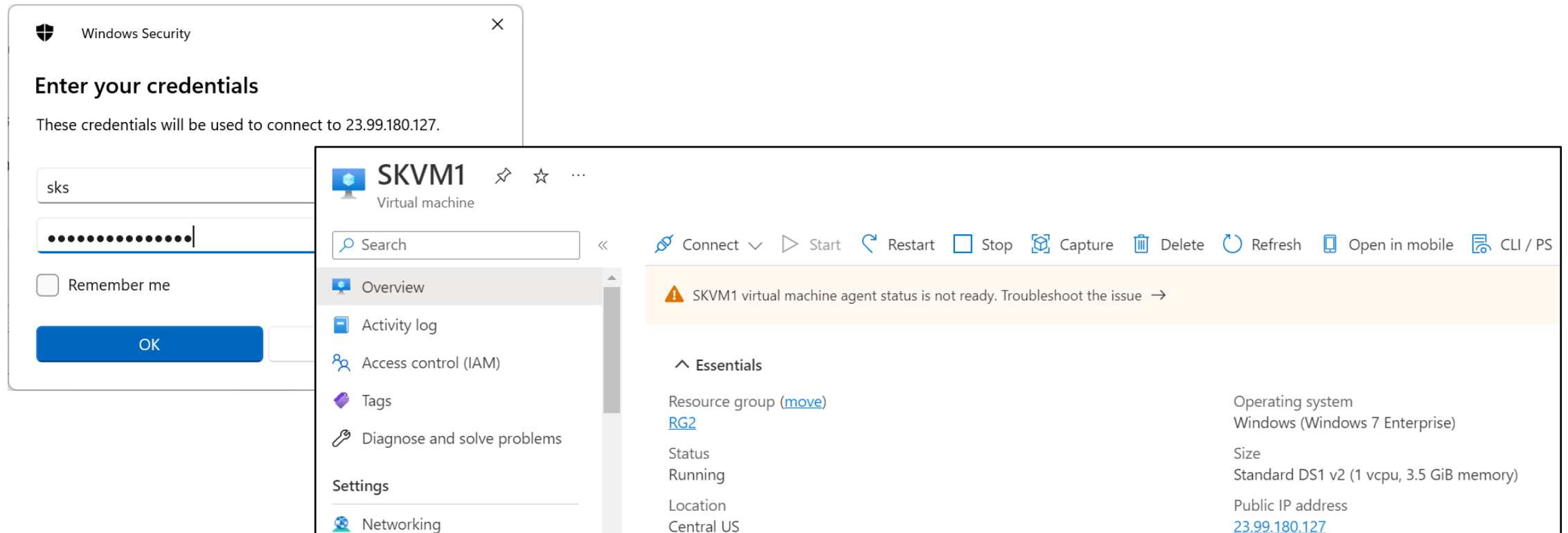
Create an Azure VM with the Azure portal

- The **Azure portal** provides an easy-to-use browser-based user interface that enables you to create and manage all your Azure resources.

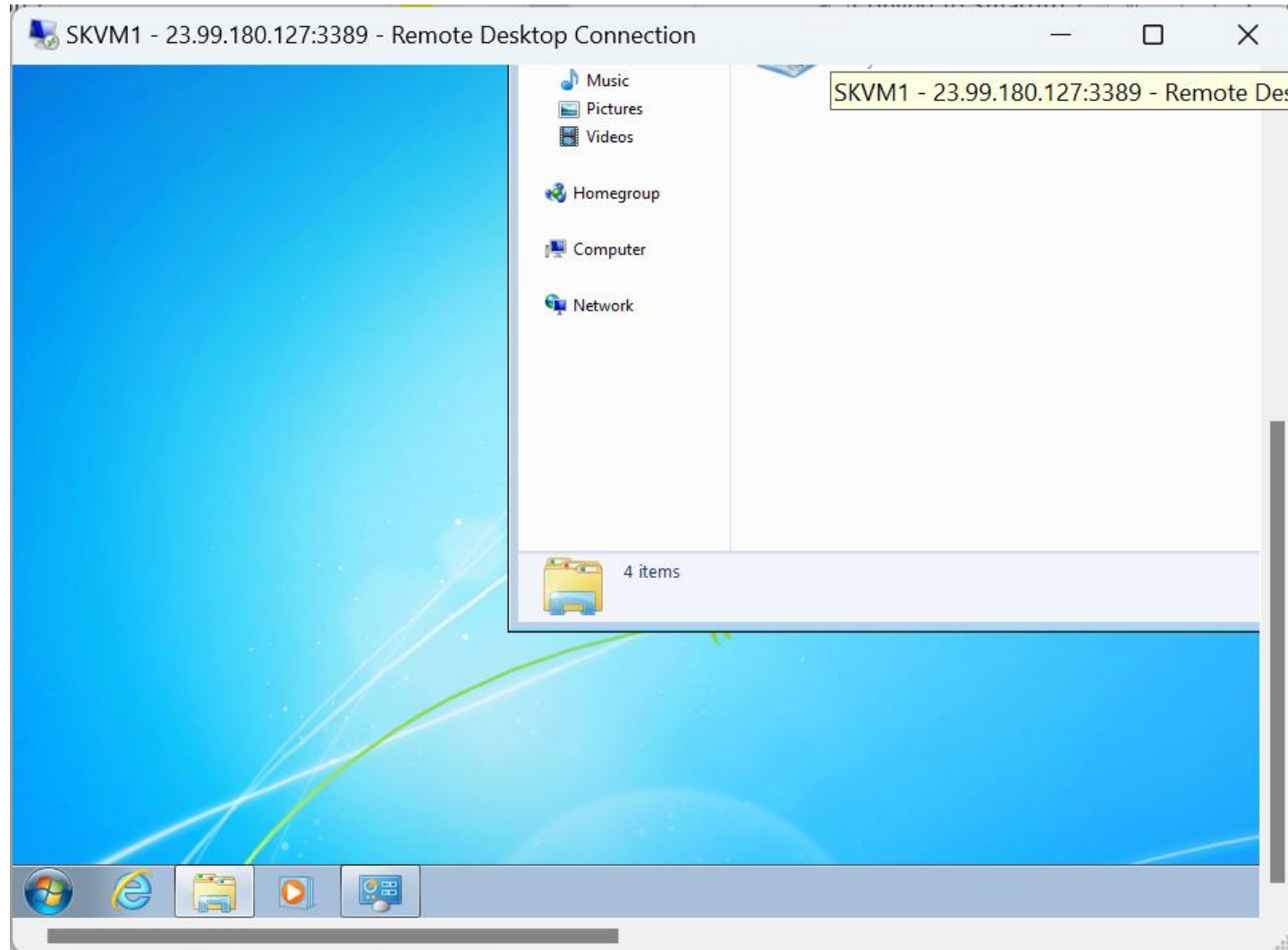


Create an Azure VM with the Azure portal

- Created Windows 7 Enterprise VM, connect with RDP



Create an Azure VM with the Azure portal



Create an Azure VM with the Azure CLI

- Login to azure portal –
 - `az login`
- Create RG
- `az group create --name myResourceGroup --location eastus`
- Create VM
- `az vm create --resource-group myResourceGroup --name myVM --image Debian --admin-username azureuser --generate-ssh-keys`

Create an Azure VM with the Azure CLI

```
C:\Users\Sharad>az vm create --resource-group myResourceGroup --name myVM --image Debian --admin-username azureuser --generate-ssh-keys
SSH key files 'C:\Users\Sharad\.ssh\id_rsa' and 'C:\Users\Sharad\.ssh\id_rsa.pub' have been generated under ~/.ssh to allow SSH access to the VM. If using machines without permanent storage, back up your keys to a safe location.
It is recommended to use parameter "--public-ip-sku Standard" to create new VM with Standard public IP. Please note that the default public IP used for VM creation will be changed from Basic to Standard in the future.
{
  "fqdns": "",
  "id": "/subscriptions/0673c648-4758-4bd8-9e12-7095bf6bfc82/resourceGroups/myResourceGroup/providers/Microsoft.Compute/virtualMachines/myVM",
  "location": "eastus",
  "macAddress": "60-45-BD-D4-84-F7",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "172.174.40.209",
  "resourceGroup": "myResourceGroup",
  "zones": ""
}
C:\Users\Sharad>
```

☐	 myVM	Virtual machine	East US
☐	 myVM_OsDisk_1_2145b31fd332479484...	Disk	East US
☐	 myVMNSG	Network security group	East US
☐	 myVMPublicIP	Public IP address	East US

Azure ARM template

Overview

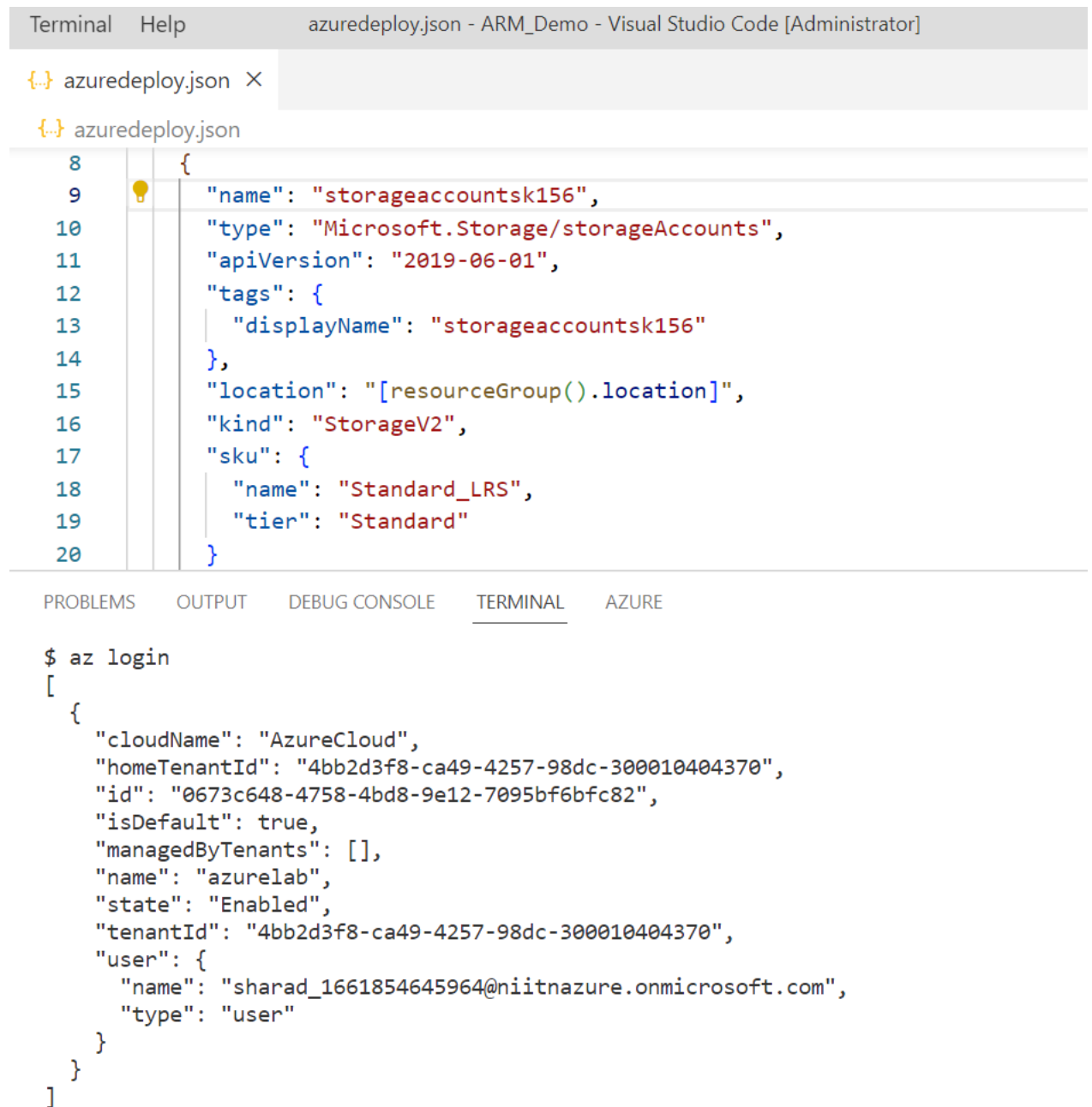
- Are you tired of clicking around the Azure portal creating virtual machines, storage accounts, and app services? Do you need a reliable method of creating the same environment multiple times? It sounds like you need to write some ARM templates!

Overview

- Azure Resource Manager (ARM) templates enable you to define the infrastructure requirements for your deployments on Azure
- ARM stands for Azure Resource Manager, which is Microsoft's deployment and management service for Azure. You use the ARM management layer for creating, updating, and deleting Azure resources

Demo

- JSON ARM templates use a declarative code syntax that you treat like application code.



The screenshot displays the Visual Studio Code interface. The top bar shows 'Terminal', 'Help', and the file name 'azuredeploy.json - ARM_Demo - Visual Studio Code [Administrator]'. The editor window shows the 'azuredeploy.json' file with the following JSON content:

```
8 {
9   "name": "storageaccountsk156",
10  "type": "Microsoft.Storage/storageAccounts",
11  "apiVersion": "2019-06-01",
12  "tags": {
13    "displayName": "storageaccountsk156"
14  },
15  "location": "[resourceGroup().location]",
16  "kind": "StorageV2",
17  "sku": {
18    "name": "Standard_LRS",
19    "tier": "Standard"
20  }
```

The bottom panel shows the 'TERMINAL' tab with the following output:

```
$ az login
[
  {
    "cloudName": "AzureCloud",
    "homeTenantId": "4bb2d3f8-ca49-4257-98dc-300010404370",
    "id": "0673c648-4758-4bd8-9e12-7095bf6bfc82",
    "isDefault": true,
    "managedByTenants": [],
    "name": "azurelab",
    "state": "Enabled",
    "tenantId": "4bb2d3f8-ca49-4257-98dc-300010404370",
    "user": {
      "name": "sharad_1661854645964@niitnazure.onmicrosoft.com",
      "type": "user"
    }
  }
]
```

Demo

```
Sharad@BLR-REG-338660 MINGW64 /d/NIIT-SR/Delivery/Azure/Sharad_Demos/ARM_Demo
$ az account set --subscription "azurelab"
```

```
Sharad@BLR-REG-338660 MINGW64 /d/NIIT-SR/Delivery/Azure/Sharad_Demos/ARM_Demo
$ az account list \
  --refresh \
  --query "[?contains(name, 'azurelab')].id" \
  --output table
```

Result

```
-----
0673c648-4758-4bd8-9e12-7095bf6bfc82
```

```
Sharad@BLR-REG-338660 MINGW64 /d/NIIT-SR/Delivery/Azure/Sharad_Demos/ARM_Demo
$ az configure --defaults group=RG1
```

Demo

```
Sharad@BLR-REG-338660 MINGW64 /d/NIIT-SR/Delivery/Azure/Sharad_Demos/ARM_Demo
$ templateFile="azuredeploy.json"
today=$(date +"%d-%b-%Y")
DeploymentName="blanktemplate-$today

az deployment group create \
  --name $DeploymentName \
  --template-file $templateFile
{
  "id": "/subscriptions/0673c648-4758-4bd8-9e12-7095bf6bfc82/resourceGroups/RG1/providers/Microsoft.Resources/deployments/blanktemplate-04-Feb-2023",
  "location": null,
  "name": "blanktemplate-04-Feb-2023",
  "properties": {
    "correlationId": "1045267d-c5b4-4cc3-983e-d2f172a5e091",
    "debugSetting": null,
    "dependencies": [],
    "duration": "PT30.5152522S",
    "error": null,
    "mode": "Incremental",
    "onErrorDeployment": null,
    "outputResources": [
      {
        "id": "/subscriptions/0673c648-4758-4bd8-9e12-7095bf6bfc82/resourceGroups/RG1/providers/Microsoft.Storage/storageAccounts/storageaccountsk156",
        "resourceGroup": "RG1"
      }
    ]
  }
}
```

Demo

Microsoft Azure

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Storage accounts

...

Stackroute

Create

Restore

Manage view

Refresh

Export to CSV

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Assign tags

Delete

Filter for any field...

Subscription equals **all**

Resource group equals **all**

Location equals **all**

Add filter

Showing 1 to 1 of 1 records.

☐ Name	Type	Kind	Resource group
☐ storageaccountsk156	Storage account	StorageV2	RG1

Thank You